T. Narikiyo
Associate Professor,
Department of Information & Control Engineering,
Toyota Technological Institute,
Tenpaku, Nagoya 468 Japan

T. Izumi
Associate Professor,
Department of Electrical Engineering,
Yamaguchi University,
Ube, Yamaguchi 755 Japan

On Model Feedback Control for Robot Manipulators

Robot manipulators are highly coupled nonlinear systems and their motions are influenced by uncertain dynamics such as coulomb friction. These nonlinearities and uncertainties disturb the performance of control systems. In this paper, a control design methodolgy is proposed for the purpose of reducing the adverse effects of parameter uncertainties and disturbances. This control structure is similar to that of classical control. Unlike classical control, this control methodology accommodates multivariable control systems with uncertain dynamics and disturbances. The control design methodology is applied to a three-degrees-of-freedom directly driven robot. Simulation and experimental results demonstrate excellent robustness.

Introduction

Robot manipulators are constructed with d-c servo motors and various other mechanical elements, and their motions are influenced by uncertain dynamics and disturbances. These unknown dynamics and parameter variations seriously disturb the performance of its control system during operations. In the control system design of robot manipulators, robustness considerations are important (Furuta et al., 1984; Seraji, 1987). In order to improve the robustness a PID control system has been introduced. In the usual PID control system, high powered actuators are required so as to ignore the robot dynamics. Recent equipment in PID control systems has become large, and the overall system is not flexible enough to permit changes of control strategy. Therefore, many control design methodologies have been proposed (Balestrino et al., 1983; Koivo et al., 1983; Reswick, 1956; Ryan, 1983; Spong, 1986). For example, the model reference adaptive control systems (MRACS) (Balestrino et al., 1983; Koivo, 1983) and a sliding mode control based on variable structure systems theory have been developed (Ryan, 1983; Spong, 1986). But there are many unsolved problems preventing implementation. Generally, a huge amount of calculations are required in order to make these control systems feasible. In the MRACS, for example, it is assumed that the nonlinearity of robot dynamics is qualitatively known and invariant during operation. In the sliding mode control, chattering of machinery is caused by the discontinuous control law.

The aim of this paper is to incorporate robustness considerations into classical control design and to improve the robot control design methodology. A new model feedback control system (MFCS) will be proposed and analyzed with close attention to robustness. It will be theoretically shown that MFCS has a high insensitivity and robustness in the scalar case. This design methodology can be easily extended to the multivariable system while maintaining robustness. In the multivariable system the classical Nyquist array methods and diagonal dominance conditions for uncertain matrices will be incorporated for accommodating the multivariable MFCS, which is influenced by uncertain dynamics. The new theoretical framework shows that the synthesis of multivariable MFCS can be attributed to that of the scalar MFCS.

The MFCS will be applied to the trajectory control of a direct-drive robot (DD robot) with three degrees of freedom. Experimental results and numerical examples show that the MFCS is easily implemented and able to reduce the adverse effects of uncertainties and parameter variations.

2 Scalar Systems

In this section, the properties of the model feedback control system shown in Fig. 1 will be investigated. We consider an additive type of uncertainty in the transfer function description of the plant

$$G_p(s) = G_0(s) + \delta G(s)$$

where $G_0(s)$ is the transfer function of the nominal approximating the true plant behavior and $G_p(s)$ is the uncertain transfer function of the true plant. Generally, it is unnecessary to distinguish between the uncertainties occurring at the input and those occurring at the output of the plant when the additive form of uncertainty is considered (Arkun et al., 1984). $G_c(s)$ is the compensator transfer function. $f(s)$ is a transfer function inserted into the inner loop in order to reduce the gain of the transfer function relationship between $r(s)$ and $u(s)$ and is arbitrarily chosen.

The main idea of this control scheme is illustrated in Fig.

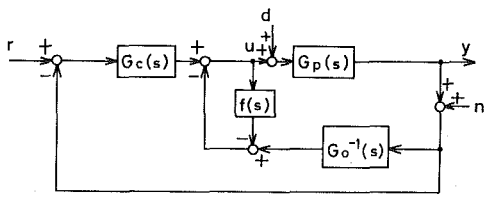


Fig. 1 Model feedback control system

Contributed by the Dynamic Systems and Control Division for publication in the JOURNAL OF DYNAMIC SYSTEMS, MEASUREMENT, AND CONTROL. Manuscript received by the Dynamic Systems and Control Division March 24, 1989, revised manuscript received October 1, 1990. Associate Editor: S. Jayasuriya.

Copyright © 1991 by ASME
Downloaded From: http://dynamicsystems.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jdsmaa/26172/ on 06/14/2017 Terms of Use: http://www.asme.org/a

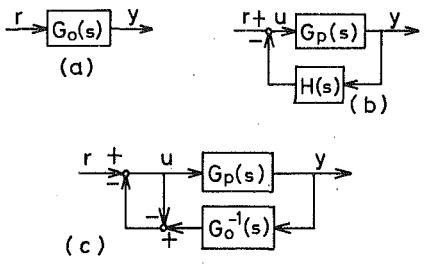


Fig. 2 Equivalent systems

2. In this figure three equivalent systems are illustrated. If $H(s)$ satisfies Eq. (1), then the input-output relation of the system illustrated in Fig. 2(*a*) is equivalent to that of the system illustrated in Fig. 2(*b*).

$$\frac{G_p(s)}{1+H(s)G_p(s)} = G_0(s) \tag{1}$$

From this condition $H(s)$ is given by

$$H(s) = G_0^{-1}(s) - G_p^{-1}(s). \tag{2}$$

Furthermore Fig. 2(*b*) gives

$$u(s) = r(s) - H(s)y(s).$$

By substituting Eq. (2) and $u(s) = G_p^{-1}(s)y(s)$ into the right-hand side of this equation, the following relation is obtained.

$$u(s) = r(s) + u(s) - G_0^{-1}(s)y(s) \tag{3}$$

This input-output relation is illustrated in Fig. 2(*c*). In this system the gain of the transfer function between $r(s)$ and $u(s)$ is infinity. Then $f(s)$ must be inserted into this loop in order to reduce the loop gain. Usually $f(s)$ is arbitrarily selected under the condition that the transfer function between $r(s)$ and $u(s)$ is stable. A similar control system was first proposed by Rezswick (1956). His techniques replaced $f(s)$ with $k(s)G_0(s)$ and $k(s)$ was arbitrarily chosen. But the inverse form $G_0^{-1}(s)$ was not used in the inner loop. Usually implementation of $G_0^{-1}(s)$ is not easy. Nevertheless, the inverse form is preferred to the form of $k(s)\ G_0(s)$ because it can be easily extended to multivariable control systems as stated later. In the MFCS, $G_0(s)$ is not a transfer function of a real system but a nominal transfer function of the plant. Therefore the transfer function $G_0(s)$ should be carefully selected so that the inverse form $G_0^{-1}(s)$ can be implemented.

In the model feedback control system illustrated in Fig. 1, $G_r(s)$, the closed-loop transfer function, $G_d(s)$, the transfer function relationship between disturbance $d(s)$ and output $y(s)$, and $S(s)$, the sensitivity function for parameter variation, are as given below:

$$G_r(s) = \frac{G_p(s)F(s)G_c(s)}{1+G_p(s)F(s)\,(G_c(s)+G_0^{-1}(s))} \tag{4}$$

$$G_d(s) = \frac{G_p(s)}{1+G_p(s)F(s)\,(G_c(s)+G_0^{-1}(s))} \tag{5}$$

$$S(s) = \frac{1}{1+G_p(s)F(s)\,(G_c(s)+G_0^{-1}(s))} \tag{6}$$

where $F(s) = 1/(1 - f(s))$ and sensitivity function is defined by

$$S(s) = \frac{(G_r(s) - G_r^{\,0}(s))/G_r^{\,0}(s)}{(G_p)(s) - G_0(s))/G_0(s)}$$

where

$$G_r^{\,0}(s) = \frac{G_0(s)F(s)G_c(s)}{1+G_0(s)F(s)\,(g_c(s)+G_0^{-1}(s))}.$$

If $f(s)$ is selected as 1, then the transfer functions are given by

$$G_r(s) = G_c(s)/(G_c(s) + G_0^{-1}(s)),\ G_d(s) = 0.$$

and the sensitivity function is given by $S(s) = 0$. This means that the property of tracking to a command input and the property of disturbance reduction and insensitivity are all improved when $f(s) = 1$. On the other hand, if the measurement noise $n(s)$ is added to the output $y(s)$, then the transfer function relationship between $n(s)$ and $y(s)$ is given by

$$G_n(s) = \frac{-G_p(s)F(s)\,(G_c(s)+G_0^{-1}(s))}{1+G_p(s)F(s)\,(G_c(s)+G_0^{-1}(s))} \tag{7}$$

and the relationship between $G_n(s)$ and $S(s)$ is given by

$$-G_n(s) + S(s) = 1. \tag{8}$$

This means that there is a trade-off between $G_n(s)$ and the sensitivity function. At low frequencies the sensitivity is re-

Nomenclature

I = unit matrix with appropriate dimension
$G_0(s)$ = transfer function (matrix) of the nominal, approximating the true plant behavior
$G_p(s)$ = uncertain transfer function (matrix) of the true plant
$\delta G(s)$ = scalar transfer function of the model-plant mismatch
$Z_+[P(s)]$ = a set of unstable poles of $P(s)$
$G_c(s)$ = transfer function (matrix) of the compensator
$f(s)$ = transfer function (matrix) of the filter
$E(s)$ = transfer function matrix of the model-plant mismatch
$N_{-1+jo(P(s))}$ = the number at which Nyquist diagram of $P(s)$ encircles the $(-1+jo)$ point clockwise
J_0 = inertia matrix of d-c servo motor
θ = vector of generalized coordinates
$\dot{\theta}$ = vector of generalized velocities ($d\theta/dt$)
$\mathbf{v}$ = vector of input voltages
$M(\theta)$ = inertia matrix of robot manipulator
$h(\theta,\dot{\theta})$ = vector of Coriolis and centrifugal forces
$g(\theta)$ = vector of gravitational forces
D = constant matrix which represents the relationship between voltages and torques
K = mixed constant matrix of back EMF and linear friction forces
$F\mathrm{sgn}\dot{\theta}$ = vector of coulomb friction forces
α = negative constant which is arbitrarily assigned
ω = angular velocity of the desired trajectory
ϕ = vector of trajectory errors
T = design parameter of the filter transfer function (matrix)
T_s = sampling time for experiments
m_i = mass of the ith element of robot manipulator
l_i = length of the ith element of robot manipulator
I_i = moment of inertia of the ith element of robot manipulator
l_{gi} = length from the (i-1)th joint to the ith center of mass

Abbreviations

MFCS = model feedback control system
MRACS = model reference adaptive control system
DD robot = direct-drive (directly driven) robot

Downloaded From: http://dynamicsystems.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jdsmaa/26172/ on 06/14/2017 Terms of Use: http://www.asme.org/a

duced, while at high frequencies the influence of the measurement noise is reduced.

In order to discuss robust stability the following must be assumed.

(*a*) If $\delta G(s) = 0$ then the closed-loop system is stable.

(*b*) Zeros of $G_0(s)$ are all in the open left half plane.

Proposition 1. The MFCS illustrated in Fig. 1 is assumed to satisfy (*a*) and (*b*). A closed loop system has robust stability if

(i) $Z_+[G_0(s)] = Z_+[G_0(s) + \delta G(s)]$

(ii) $|1 + H(j\omega)| > |\delta G(j\omega) G_0^{-1}(j\omega) H(j\omega)| \qquad \omega \epsilon R$

where $Z_+[P(s)]$ is a set of unstable poles of $P(s)$ and

$$H(s) = F(s)(1 + G_0(s)G_c(s)).$$

This proposition is easily proved. The characteristic polynomial equation is given by

$$1 + G_p(s)F(s)(G_c(s) + G_0^{-1}(s)) = 0$$

By substituting $G_p(s) = G_0(s) + \delta G(s)$ into this equation, we obtain

$$1 + H(s) + \delta G(s) G_0^{-1}(s) H(s) = 0.$$

If the conditions (i, ii) are satisfied, the number of encirclements of the $(-1 + j0)$ point clockwise by the Nyquist diagram of $H(s) + \delta G(s) G_0^{-1}$ is the same number as for $H(s)$. Therefore, on the basis of the Nyquist stability criterion (Rosenbrock, 1974), it is easily shown that these conditions are sufficient for robust stability of the closed-loop system.

Now the conditions (i) and (ii) can be replaced with the following conditions.

Proposition 2. The MFCS illustrated in Fig.1 is assumed to satisfy (*a*) and (*b*). The closed-loop system has robust stability if

(i) Zeros of $(G_0(s) + \delta G(s))$ are in the open left half plane.

(ii) $|1 + G_0(j\omega)G_c(j\omega)| > |F^{-1}(j\omega)(1 + \delta G(j\omega) G_0^{-1}(j\omega))^{-1}|$, $\omega \epsilon R$

Proof The closed-loop transfer function is rewritten as

$$G_r(s) = \frac{G(s)}{1 + G(s) + F^{-1}(s)(1 + \delta G(s) G_0^{-1}(s))^{-1}},$$

where $G(s) = G_0(s) G_0(s)$.

If $Z_+[F^{-1}(s)(1 + \delta G(s) G_0^{-1}(s))^{-1}] \subset C^-$ (open left half plane) then a sufficient condition for stability of the closed-loop system is given by

$$|1 + G(j\omega)| > |F^{-1}(j\omega)(1 + \delta G(j\omega) G_0^{-1}(j\omega))^{-1}|.$$

On the other hand, $F^{-1}(s)(1 + \delta G(s) G_0^{-1}(s))^{-1}$ can be rewritten as

$$F^{-1}(s) G_0(s) (G_0(s) + \delta G(s))^{-1}.$$

Therefore, if condition (i) is satisfied, then $Z_+[F^{-1}(s)(1 + \delta G(s) G_0^{-1}(s))^{-1}] \subset C^-$.

3 Multivariable Systems

The MFCS illustrated in Fig. 1 is easily extended to multivariable systems. But the discussion of the multivariable MFCS is slightly different from that of the scalar case. In the multivariable system, the classical Nyquist array methods and diagonal dominance conditions for uncertain matrices are incorporated to accommodate multivariable MFCS having uncertain dynamics. The new theoretical framework shows that the synthesis of the multivariable MFCS can be attributed to that of the scalar MFCS.

The multivariable MFCS is also illustrated in Fig. 1. The command input vector to the system is denoted by $r(s)$, the disturbance vector by $d(s)$ and the measurement noise vector by $n(s)$. The plant transfer function matrix is denoted by the $n \times n$ rational matrix $G_p(s)$. Compensator $G_c(s)$, $f(s)$ and model transfer function matrix $G_0(s)$ are all $n \times n$ diagonal matrices whose elements are rational function of s. These are denoted by

$$G_c(s) = \text{diag}\,[g_1^c(s),\ldots,g_{nc}(s)]$$
$$G_0(s) = \text{diag}\,[g_1^0(s),\ldots,g_n^0(s)]$$
$$f(s) = \text{diag}\,[f_1(s),\ldots,f_n(s)].$$

Then the transfer function matrix relationship between $y(s)$ and $r(s)$ is given by

$$W(s) = G_p(s)[1 - f(s) + (G_c(s) + G_0^{-1}(s))G_p(s)]^{-1} G_c(s). \quad (9)$$

Letting $[I - f(s)]^{-1} = (1/k(s))I$, namely

$$f(s) = \text{diag}\,[1 - k(s),\ 1 - k(s),\ldots,1 - k(s)]$$

then the closed-loop transfer function matrix $W(s)$ can be written as

$$W(s) = [1 + G_c^{-1}(s)(k(s)G_p^{-1}(s) + G_0^{-1}(s))]^{-1}. \quad (10)$$

Also in the multivariable system an additive type of uncertainty is considered.

$$G_p(s) = G_0(s) + E(s)$$

$G_0(s)$ is defined above. $E(s)$ is the model-plant mismatch of the error component. Then

$$W(s) = [1 + G(s) + k(s)\Delta G(s)]^{-1} G(s) \quad (11)$$

where $\Delta G(s) = [I + E(s) G_0^{-1}(s)]^{-1}$, $G(s) = G_0(s) G_c(s)$. Thus

$$G(s) = \text{diag}\,[g_1(s),\ldots,g_n(s)],\ g_i(s) = g_i^0(s) g_i^c(s).$$

In eq. (11), $k(s)\,\Delta G(s)$ can be considered as parameter variations similar to $\delta G(s)$ appearing in the scalar MFCS. Hence, discussion of robust stability can be done with the new theoretical framework, but in order to do this the following definitions are required.

$$\underline{\sigma}(I + G(j\omega)) = \min_{i,\omega} |1 + g_i(j\omega)| \quad (12)$$

$$\bar{\sigma}(\Delta G(j\omega)) = \max_{i,\omega} \sum_{k=1}^{n} |\Delta g_{ik}(j\omega)| \quad (13)$$

where $\bar{\sigma}(.)$ is a matrix norm induced by the vector norm, also $\bar{\sigma}(.)$ satisfies the following equations.

(i) $\bar{\sigma}(P^{-1}) \geq [\bar{\sigma}(P)]^{-1}$

(ii) $[\bar{\sigma}(P^{-1})]^{-1} = \min_{x \neq 0} \dfrac{\|Px\|}{\|x\|}$, $x = (x_1,\ldots,x_n)^t$

$$\|x\| = \max_i |x_i|$$

By these definitions, dominance conditions for uncertain matrices can be give as follows.

Proposition 3. Let $G(s)$, $\Delta G(s)$ be the complex matrix defined above.

If

$$|k(j\omega)| < \underline{\sigma}(I + G(j\omega))/\bar{\sigma}(\Delta G(j\omega)) \quad (14)$$

then the complex matrix $I + G(s) + k(s)\,\Delta G(s)$ is diagonally row dominant.

Proof

$$|1 + g_i(j\omega)| - |k(j\omega)\Delta g_{ii}(j\omega)| - \left|\sum_{l \neq i} k(j\omega)\Delta g_{il}(j\omega)\right|$$

$$> |1 + g_i(j\omega)| - \frac{\underline{\sigma}(I + G(j\omega))}{\bar{\sigma}(\Delta(j\omega))} |\Delta g_{ii}(j\omega)| - \frac{\bar{\sigma}(I + G(j\omega))}{\bar{\sigma}(\Delta G(j\omega))} \sum_{l \neq i} |\Delta g_{il}(j\omega)|$$

Downloaded From: http://dynamicsystems.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jdsmaa/26172/ on 06/14/2017 Terms of Use: http://www.asme.org/a

$$> |1+g_i(j\omega)| - \frac{\underline{\sigma}(I+G(j\omega))}{\bar{\sigma}(\Delta G(j\omega))} |\Delta g_{ii}(j\omega)| - \frac{\underline{\sigma}(I+G(j\omega))}{\bar{\sigma}(\Delta G(j\omega))} [\bar{\sigma}(\Delta G(j\omega)) - |\Delta g_{ii}(j\omega)|] = |1+g_i(j\omega)| - \underline{\sigma}(I+G(j\omega)) \qquad \omega, i=1,\ldots,n$$

Thus

$$|1+g_i(j\omega)| - |k(j\omega)\Delta g_{ii}(j\omega)| > |\sum_{l\neq i} k(j\omega)\Delta g_{il}(j\omega)| \qquad \omega,\ i=1,\ldots,n$$

$$|1+g_i(j\omega)+k(j\omega)\Delta g_{ii}(j\omega)| > \sum_{l\neq i} k(j\omega)\Delta g_{il}(j\omega)| \qquad \omega,\ i=1,\ldots,n \quad (15)$$

The above proposition and the following Lemma allow one to deduce conditions for robust stability of the uncertain multivariable system.

(Lemma) Stability of Diagonal Dominance Matrix (Rosenbrock, 1974)

Let $G(s)$ be an open-loop transfer function matrix.

Let $I+G(s)$, the return difference matrix, be diagonally dominant. Then, the closed-loop system is stable if and only if

$$\sum_{i=1}^{n} N_{-1+j0}(g_{ii}(s)) = -P_0.$$

Where p_0 is the number of open-loop poles in the closed right half plane and $N_{-1+j0}(g(s))$ is the number of clockwise encirclements of the $(-1+j_0)$ point by the Nyquist diagram of $g(s)$.

The characteristic equation of the MFCS is given by

$$\det\,[I+G(s)+k(s)\Delta G(s)]=0$$

where $G(s)$, $\Delta G(s)$ are defined in Eq. (11). Stability of the closed-loop system is evaluated by the location of zeros of the characteristic equation. But if $I + G(s) + k(s)\ \Delta G(s)$ is diagonal dominant, then the stability of the MFCS can be checked by the following proposition.

Proposition 4. Let $G(s)$, $\Delta G(s)$ be complex matrices defined above.

If

$$\text{(i)}\ |k(j\omega)| < \underline{\sigma}(I+G(j\omega))/\bar{\sigma}(\Delta G(j\omega))$$

$$(ii)\ \sum_i N_{-1+j0}(g_{ii}(s)) = -p_0$$

then the closed-loop system is stable.

Proof Condition (i) shows that $I+G(s)+\Delta G(s)$ is diagonally row dominant. Then by Proposition 3 the following inequality is derived.

$$|1+g_{ii}(j\omega)| > |k(j\omega)\Delta g_{ii}(j\omega)| \quad \omega,\ i=1,\ldots,n$$

This means that the following relation is satisfied.

$$N_{-1+j0}(g_{ii}(s)) = N_{-1+j0}(g_{ii}(s)+k(s)\Delta g_{ii}(s)) \quad i=1,\ldots,n$$

Since the stability condition (ii) holds

$$\sum_i N_{-1+j}(g_{ii}(s)+k(s)\Delta g_{ii}(s)) = -p_0.$$

Now by the above Lemma it can be proved that the closed-loop system has robust stability.

Diagonal dominance plays an important rote in multivariable control system design. It was first reported by Rosenbrock (1974) that the Nyquist stability criterion and diagonal dominance can be incorporated to cover the case of parameter uncertainty. But no one has yet considered how to realize the diagonal dominance. On the other hand, diagonal dominance can be easily given by MFCS, because in the multivariable system the filter $f(s)$ can be arbitrarily selected to give diagonal dominance.

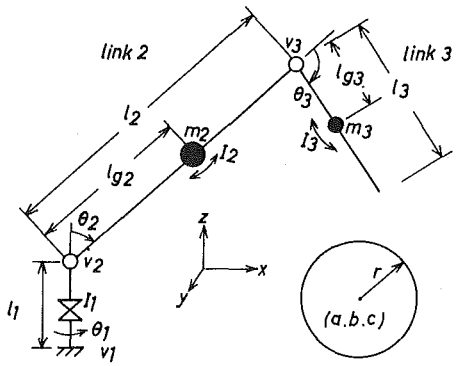


Fig. 3 Schematic model of three-degrees-of-freedom robot

Theoretically, $G_0(s)$ can always be selected as a diagonal transfer function matrix. As stated in proposition 3 and proposition 4, filter $f(s)$ plays an important role in order to realize diagonal dominance. If $f(s)$ is selected as I in eq. (11), then the closed-loop transfer function matrix $W(s)$ is given by

$$W(s) = [\mathrm{I}+G(s)]^{-1}G(s).$$

This means that, when $f(s)$ is I, the transfer function matrix of a plant equals $G_0(s)$. But it may happen that $f(s)$ puts too great a burden on actuators. In this case, $G_0(s)$ or the filter $f(s)$ should be changed in order to accommodate the MFCS. How to mediate between $G_0(s)$ and $f(s)$ is beyond the scope of this paper and it remains an open problem.

4. Application to Robot Manipulators

4.1 Numerical Example. Here we consider a robust trajectory control system. Equations of motion of the robot manipulators are highly coupled and are nonlinear systems. Figure 3 shows a schematic model of the three-degrees-of-freedom robot. Equations of motion for this robot are given by

$$[J_0+M(\theta)]\ddot{\theta}+h(\theta,\dot{\theta})+\mathrm{g}(\theta)+\mathrm{K}\dot{\theta}+F\mathrm{sgn}\,\dot{\theta}=D\mathbf{v} \quad (16)$$

where $\theta = [\theta_1,\ \theta_2,\ \theta_3]^t$ denotes the vector of generalized coordinates and $\mathbf{v} = [v_1,\ v_2,\ v_3]^t$ denotes the vector of input voltages supplied to the d-c servo motors. Each of these matrices is given by

$$J_0=\mathrm{diag}[J_{11},J_{22},J_{33}], h(\theta,\dot{\theta}=[h_1,h_2,h_3]^t$$

$$\mathrm{g}(\theta)=[g_1,\ g_2,\ g_3]^t,\ K=\mathrm{diag}[K_1,\ K_2,\ K_3]$$

$$F=\mathrm{diag}\ [F_1,\ F_2,\ F_3],\ D=\mathrm{diag}\ [D_1,\ D_2,\ D_3]$$

$$M(\theta)=[M_{ij}(\theta)]\ ,\ (i,\ j=1,\ 2,\ 3)$$

$$M_{11}(\theta)=I_1+m_2 1g_2{}^2\sin\theta_2+m_3(1_2\sin\theta_2+1g_3\sin(\theta_2+\theta_3))^2$$

$$M_{12}(\theta)=M_{21}(\theta)=M_{13}(\theta)=M_{31}(\theta)=0$$

$$M_{22}(\theta)=I_2+m_2 1g_2^2+m_3 I_2^2+I_3+m_3 1g_3^2+2m_3 1_2 1g_3\cos\theta_3$$

$$M_{23}(\theta)=I_3+m_3 1g_3^2+m_3 1_2 1g_3\cos\theta_3$$

$$M_{32}(\theta)=M_{23}(\theta)$$

$$M_{33}(\theta)=I_3+m_3 1g_3{}^2$$

$$h_1(\theta,\dot{\theta})=2m_2 1g_2{}^2\sin\theta_2\cos\theta_2+2m_3(1_3\sin\theta_2+1g_3\sin(\theta_2+\theta_3))1_2\cos\theta_2+2m_3(1_2\sin\theta_2+1g_3\sin(\theta_2+\theta_3))\times 1g_3\cos(\theta_2+\theta_3)\theta_1\theta_2+2m_3(1_2\sin\theta_2+1g_3\sin(\theta_2+\theta_3))1g_3\cos(\theta_2+\theta_3)\dot{\theta}_1\dot{\theta}_3$$

$$h_2(\theta,\dot{\theta})=-2m_3 1_2 1g_3\sin\theta_3\dot{\theta}_2\dot{\theta}_3-m_3 1_2 1g_3\sin\theta_3\theta_3{}^2-m_2 1g_2{}^2\sin\theta_2+m_3(1_2\sin\theta_2+1g_3\sin(\theta_2+\theta_3))(1_2\cos\theta_2+1g_3\cos(\theta_2+\theta_3))\dot{\theta}_1{}^2$$

$$h_3(\theta,\dot{\theta})=-m_3(1_2\sin\theta_2+1g_3\sin(\theta_2+\theta_3))\times 1g_3\cos(\theta_2+\theta_3)\dot{\theta}_1{}^2+m_3 1_2 1g_3\sin\theta_3\dot{\theta}_2{}^2$$

Downloaded From: http://dynamicsystems.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jdsmaa/26172/ on 06/14/2017 Terms of Use: http://www.asme.org/a

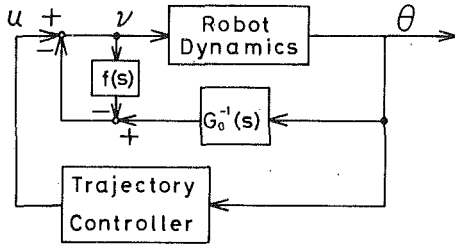


Fig. 4 Control system of robust tracking of desired trajectory

$g_1(\theta) = 0$

$g_2(\theta) = -(m_2 l g_2 + m_3 l l_2) g \sin\theta_2 - m_3 l g_3 g \sin(\theta_2 + \theta_3)$

$g_3(\theta) = -m_3 l g_3 g \sin(\theta_2 + \theta_3)$, g: acceleration of gravity.

In these equations gravitational forces and K and D can be identified accurately. Then gravitational forces can be cancelled by

$$\mathbf{v} = D^{-1}(\tau + g(\theta)). \tag{17}$$

Substituting (17) into (16) gives

$$[J_0 + M(\theta)]\ddot{\theta} + h(\theta, \dot{\theta}) + \mathbf{K}\dot{\theta} + F\mathrm{sgn}\,\dot{\theta} = \tau \tag{18}$$

One way to design a complete trajectroy control system is to obtain an exact dynamic model of the robot manipulator given in eq. (18). Unfortunately, this is impossible. Even if the physical parameters, such as masses and inertias, are given, the exact identification of robot dynamics is extremely difficult.

Some parameter identification schemes for robot manipulators have been proposed (Kwasaki et al., 1986; Mayeda,), but they require precise experiments. Therefore identified equations of motion are not equal to real equations of motion of the robot manipulator in a strict sense. On the other hand, even if any detailed identification can be done, a model-plant mismatch is inevitable. Therefore, whenever the robot manipulators are controlled based on the mathematical model of the robot, it is necessary to take the robustness into consideration.

The MFCS is applied to the control of robot manipulators. Figure 4 shows a robust tracking control scheme. In this control scheme robot dynamics is completely unknown. But we may assume that the known terms are already cancelled by the nonlinear precompensators. Then it may be expressed by an equation similar to Eq. (18). In this simulation Eq. (18) is assumed to be the equation of motion of the real system. Also, it is assumed that the mathematical model of the robot manipulator is given by

$$K\dot{\theta} = \mathbf{u} \tag{19}$$

Then the inertia forces and Coriolis and centrifugal forces are all neglected. These forces can be considered to be disturbances and parameter variations. Influences of these uncertainties may cause tracking errors or compromise stability.

In order to reduce the tracking errors, the MFCS is applied to the trajectroy control of robot manipulators. Table 1 shows parameters of the robot and desired trajectories.

As illustrated in Fig. 3, the desired trajectory is a circular movement on a vertical plane with angular velocity ω. End-effector coordinates of the robot manipulator must be traced

$$X = a + r\sin(\omega t)$$
$$Y = b$$
$$Z = c + r\cos(\omega t)$$

where $[X, Y, Z]^t$ is a vector of end-effector coordinates, $[a, b, c]^t$ is the vector of coordinates of the center of the circle, and r is the radius of the circle, and ω is the angular velocity along the trajectory.

For the linear system written by Eq. (19), the trajectory control $\mathbf{u}(t)$ is given by

Table 1 Parameters used for simulation and experiments

Fig. No.	Feedback Gain α	Angular Velocity ω Radius r	Robot Parameters	Model
5	$\alpha = -10.0$	$\omega = 1.0$ (rad/sec) $r = 0.15$(m)	$m_1 = 2.0$(kg) $m_2 = 1.2$(kg) $m_3 = 1.0$(kg) $I_1 = 0.01$(kgm^2) $I_2 = 0.01$(kgm^2) $I_3 = 0.01$(kgm^2) $lg_2 = 0.2$(m) $lg_3 = 0.3$(m) $l_2 = 0.5$(m) $l_3 = 0.5$(m) $k_1 = k_2 = k_3 = 10.0$ (kg/sec)	$K\dot{\theta} = u$ $k_1 = k_2 = k_3 = 10.0$(kg/sec)
7	$\alpha = -0.8$	$\omega = 0.2$ (rad/sec) $r = 0.05$(m)	$l_1 = 0.78$(m) $l_2 = 0.30$(m) $l_3 = 0.20$(m)	$K\dot{\theta} = u$ $k_1 = 15.5$(kg/sec) $k_2 = 15.0$(kg/sec) $k_3 = 20.0$(kg/sec)
8	$\alpha = -0.8$	$\omega = 0.5$ (rad/sec) $r = 0.05$(m)	$l_1 = 0.78$(m) $l_2 = 0.30$(m) $l_3 = 0.20$(m)	$K\dot{\theta} = u$ $k_1 = 15.5$(kg/sec) $k_2 = 15.0$(kg/sec) $k_3 = 20.0$(kg/sec)

$$\mathbf{u}(t) = [JJ(\theta)\mathrm{K}^{-1}]^{-1}[\alpha\phi - \mathrm{Q(t)}] \tag{20}$$

where α is a negative constant which is arbitrarily assigned (Freund, 1982; Narikiyo et al., 1984).

$$JJ(\theta) = \begin{bmatrix} -p(\theta)\sin\theta_1 & l_2\cos\theta_1\, Z(\theta) & l_3\cos\theta_1\cos(\theta_2+\theta_3) \\ p(\theta)\cos\theta_1 & l_2\sin\theta_1\, Z(\theta) & l_3\sin\theta_1\cos(\theta_2+\theta_3) \\ 0 & -l_2\sin\theta_2 - l_3\sin(\theta_2+\theta_3) & -l_3\sin(\theta_2+\theta_3) \end{bmatrix}$$

$$p(\theta) = l_2\sin\theta_2 + l_3\sin(\theta_2+\theta_3)$$

$$K = \mathrm{diag}[1/k_1,\ 1/k_2,\ 1/k_3]$$

$$\phi(\theta,t) = \begin{bmatrix} X(\theta) - a - r\sin(\omega t) \\ Y(\theta) - b \\ Z(\theta) - c - r\cos(\omega t) \end{bmatrix} \quad Q(t) = \begin{bmatrix} -r\omega\cos(\omega t) \\ 0 \\ r\omega\sin(\omega t) \end{bmatrix}$$

$$\mathrm{X}(\theta) = p(\theta)\cos\theta_1$$

$$Y(\theta) = p(\theta)\sin\theta_1$$

$$\mathrm{Z}(\theta) = l_2\cos\theta_2 + l_3\cos(\theta_2+\theta_3)$$

If Eq. (19) has no unmodeled dynamics, then Eq. (20) gives an exact trajectory-tracking that is commensurate with zero tracking error. But Eq. (19) denotes the model dynamics of the real system and has unmodeled dynamics. Now for robust trajectory-tracking, the MFCS is used in an inner loop. In the control scheme illustrated in Fig. 4, the model transfer function, $G_0(s)$, is given by Eq. (19):

$$G_0(s) = \mathrm{diag}\ [1/(k_1 s),\ 1/(k_2 s),\ 1/(k_3 s)]$$

And $f(s)$ is given by

$$f(s) = \mathrm{diag}\ [T/(s+T),\ T/(s+T),\ T/(s+T)].$$

The imput voltages sent to the actuators of the robot manipulator are given by

$$\mathbf{v} = \mathbf{u} - K\dot{\theta} + T\int(\mathbf{u} - K\dot{\theta})\,\mathrm{d}t \tag{21}$$

where $\mathbf{u}$, K, and T are given by Eq. (20).

In this control law, if $\mathbf{v} = \mathbf{u}$ then the tracking control is performed without the MFCS.

Figure 5 gives the simulation results of this control system. The solid lines show error trajectories between end-effector coordinates and desired trajectories with the MFCS. Broken

Downloaded From: http://dynamicsystems.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jdsmaa/26172/ on 06/14/2017 Terms of Use: http://www.asme.org/a

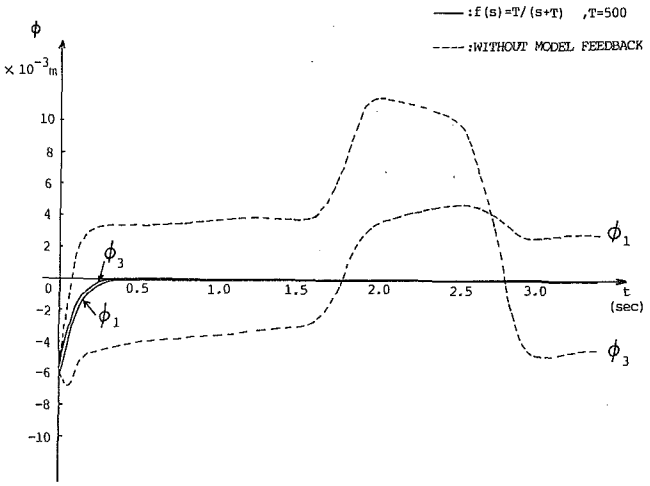


Fig. 5 Simulation result of tracking control

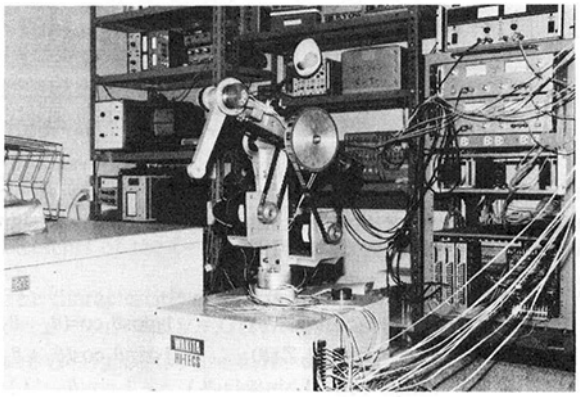


Fig. 6 Three-degrees-of-freedom robot

lines indicate error trajectories between end-effector coordinates and desired trajectories without the MFCS. In the latter the error trajectories are oscillatory and tend to increase. ϕ_1, ϕ_3 were already defined above. In this figure, ϕ_2 is not traced because the Y coordinate is always fixed at 0.

4.2 Experimental Examples. Figure 6 shows the robot manipulator used for the experiments. This robot was newly constructed for experiments. The gravitational force is approximately cancelled by the counter balance.

Joint angles are detected by potentiometes and are sent to a minicomputer for calculating the command signals. The sampling time is fixed at 16ms. The schematic model of this robot is illustrated in Fig. 3, and the equations of motion are given by Eq. (16).

Exact identification of robot parameters is very difficult, as stated previously. The identification scheme proposed by Kawasaki et al. (1986) and Mayeda (1984) gives the following equations of motion.

$$\begin{bmatrix} 3.0 & * & * \\ * & 2.5 & * \\ * & * & 0.8 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \\ \ddot{\theta}_3 \end{bmatrix} + \begin{bmatrix} 15.5 & * & * \\ * & 15.0 & * \\ * & * & 20.0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix} + \begin{bmatrix} 0 \\ -0.0.3\sin\theta_2 - 0.35\sin\theta_3 \\ -0.15\sin\theta_3 \end{bmatrix} + \begin{bmatrix} 0.2\ \mathrm{sgn}\,\dot{\theta}_1 \\ 0.08\ \mathrm{sgn}\,\dot{\theta}_2 \\ 0.35\ \mathrm{sgn}\,\dot{\theta}_3 \end{bmatrix} + \begin{bmatrix} * \\ * \\ * \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \quad (22)$$

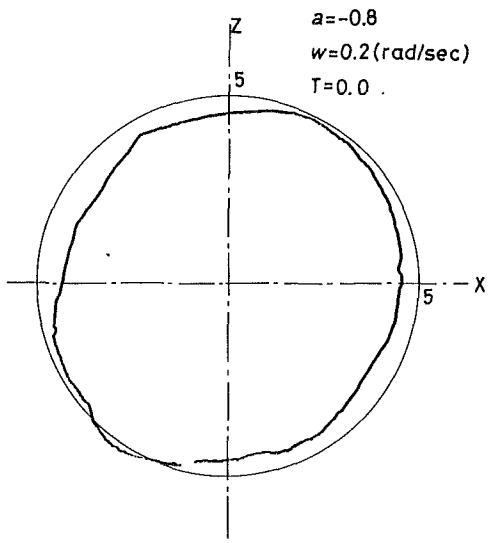


Fig. 7(*a*) Experimental result of tracking control without MFCS

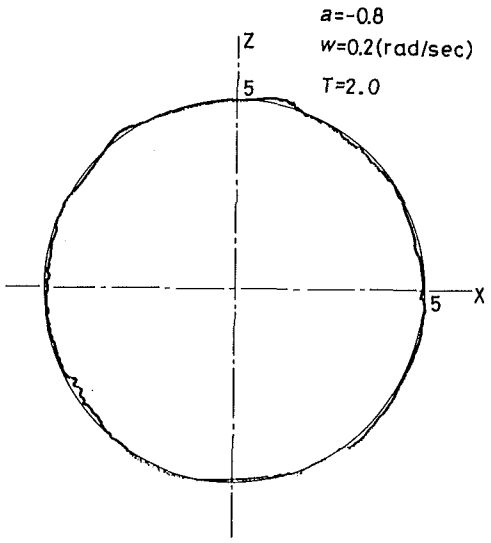


Fig. 7(*b*) Experimental result of tracking control with MFCS($T=2.0$)

In Eq. (22) * means a small value or a coupled nonlinear term which cannot be detected. Nevertheless, intrinsically nonlinear terms exist, but it is very difficult to distinguish their input-output properties from those of the others. Thus, in the model equations of motion (*)'s are replaced by zeros. In our experiments, a robust tracking control system is effected by a minicomputer system (Toshiba DS 600/40). All control parameters are shown in Table 1.

Figure 7 (*a*) shows the performance of this control system without the MFCS. The curvilinear configuration is described by the pen equipped on the end-effector of the robot manipulator itself, and the fine stroked line is a trace of the desired trajectory.

Performance of this control system is estimated by

$$I_p = \sum_{k=0}^{N} [(x(\omega k T_s) - a - r\sin(\omega k T_s))^2 + (Z(\omega k T_s) - c - r\cos(\omega k T_s))_2]$$

where T_s is sampling time and $N = 2\pi/\omega T_s$.
Settling the sampling time $T_s = 0.016$ (s) and the angular velocity $\omega = 0.2$ give

$$I_p = 0.1456296.$$

Figure 7 (*b*) shows the performance when the MFCS is ap-

Downloaded From: http://dynamicsystems.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jdsmaa/26172/ on 06/14/2017 Terms of Use: http://www.asme.org/al

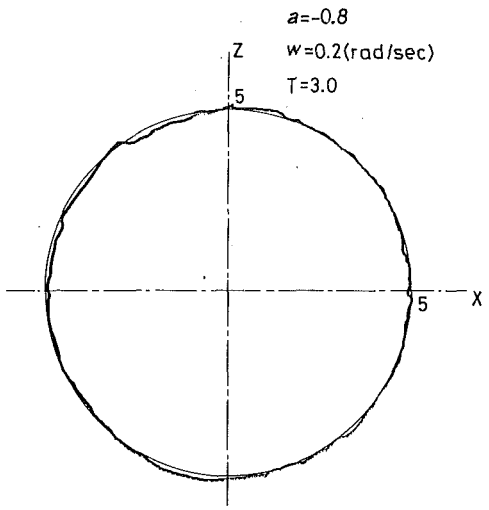


Fig. 7(c) Experimental result of tracking control with MFCS($T = 3.0$)

plied. The control input is given by Eq. (21). In this case T is set at 2.0 and the other parameters are as shown in Table 1. The trajectory error in this case is given by

$$I_p = 0.4825857 \times 10^{-2}.$$

In Fig. 7(*c*), T is set at 3.0. In this experiment the error index I_p is

$$I_p = 0.2852761 \times 10^{-2}.$$

Figure 8 shows tracking motions at a tracking speed of 0.5(rad/s). Other parameters are shown in Table 1. Figure 8(*a*) shows the performance when T is assigned 0.5. In this case the performance index is given by

$$I_p = 0.2339794 \times 10^{-1}.$$

Similarly, Fig. 8(*b*) indicates the performance when T is 2.0. In this case, the performance index is given by

$$I_p = 0.6554473 \times 10^{-2}.$$

Through these experiments the feedback gain α is fixed at -0.8 and sampling time T_s is fixed at 16ms. When $|\alpha|$ is larger than 0.8, the motion of the closed-loop system becomes unstable. Therefore $|\alpha|$ can not be larger than 0.8 also for the case of Fig. 7(*a*).

Results of these experiments show that the MFCS can reduce the tracking error caused by uncertainties, and large the value of T the better the performance. Theoretically, the closer the modulus of $T/(s + T)$ is to 1, the better the performance. But input voltages may be saturated or measurement noise may be amplified too much when the modulus of $T/(s + T)$ is close to 1. Hence, the modulus of $T/(s + T)$ cannot be assigned too close to 1. Usually the decision of $f(s)$ depends on the uncertainties and the nonlinearities. As stated earlier, at low frequencies the sensitivity should be reduced and at high frequencies the influence of measurement noise should be reduced. In order to do this at low frequencies ($s \to 0$), the modulus of $f(s)$ should be close to 1, while at high frequencies ($s \to \infty$) the modulus of $f(s)$ should be close to 0. Therefore, it is reasonable that $f(s)$ was selected as $T/(s + T)$. Nevertheless, the manner in which $f(s)$ should be adjusted to specifications remains a problem for future study.

Conclusions

A design technique for uncertain multivariable systems has been proposed. The proposed method was called the Model Feedback Control System (MFCS) and was discussed with close attention to robustness. Robust stability theorems were developed and in the case of multivariable uncertain systems the classical diagonal dominance conditions were incorporated into the MFCS.

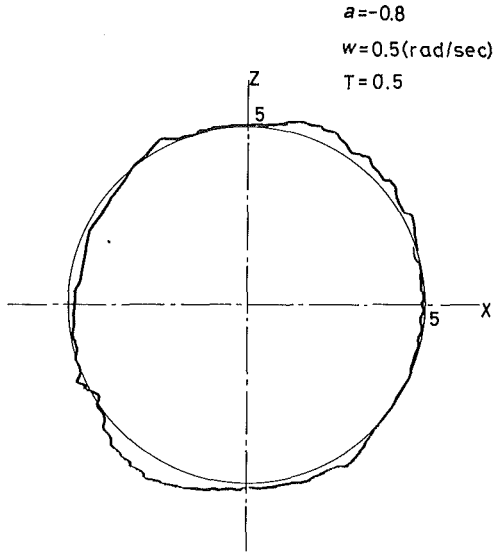


Fig. 8(*a*) Experimental result of tracking control with MFCS($T = 0.5$)

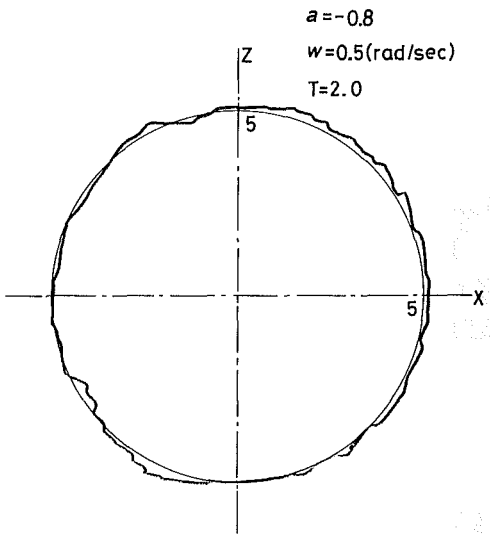


Fig. 8(*b*) Experimental result of tracking control with MFCS($T = 2.0$)

This control design methodology can be easily implemented by a digital or analog computer. This methodology is not only simple but has excellent robustness properties. When a digital computer is used as the controller, the number of calculations required in the implementation of the MFCS is nearly equal to the number of calculations required for the PID controller. Nevertheless, robustness properties of the MFCS are excellent as evidenced by the numerical and experimental results. In the experiments, MFCS was applied to a directly-driven robot with three-degrees-of-freedom. The motions of this robot are described by the coupled nonlinear equations which are difficult to identify. The application of the MFCS gave sufficiently robust motions without exact identification of the robot dynamics.

One of the remaining problems in this design methodology is how the model transfer function $G_0(s)$ and the filter transfer function $f(s)$ are to be determined.

Acknowledgments

The research described in this paper was performed at the Government Industrial Research Institute, Kyushu. The au-

Downloaded From: http://dynamicsystems.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jdsmaa/26172/ on 06/14/2017 Terms of Use: http://www.asme.org/a

thors are grateful to the associate editor, Dr. Suhada Jayasuriya, and the reviewers for their encouraging and helpful comments.

References

Arkun, Y., et al., 1984, "Robust Nyquist Array Methodology: A New Theoretical Framework for Analysis and Design of Robust Multivariable Feedback System," *Int. J. Control*, Vol. 40, No. pp. 603–629.

Balestrino A., et al., 1983, "An Adaptive Model Following Control for Robotic Manipulators," ASME JOURNAL OF DYNAMIC SYSTEMS, MEASUREMENT, AND CONTROL, Vol. 105, pp. 143–151.

Freund, E., 1982, "Fast Nonlinear Control with Arbitrary Pole Placement for Industrial Robots and Manipulators," *Int. J. Robotics Research*, Vol. 1, No. 1, pp. 65–78.

Furuta, K., et al., 1984, "Robust Control of a Robot Manipulator with Nonlinearity," *Robotica*, Vol. 2, pp. 75–81.

Kawasaki, H., et al., 1986, "Parameter Identification of Mechanical Manipulators," *Tr. SICE Jpn*, Vol. 22, No. 1, pp. 76–83 (in Japanese).

Koivo, A., et al., 1983, "Adaptive Linear Controller for Robotic Manipulators, *IEEE Tr. Automatic Control*, Vol. AC-28, No. 2, pp. 162–171.

Mayeda, H., et al., 1984, "A New Identification Method for Serial Manipulator Arms," 9-th IFAC Congress, Budapest.

Narikiyo, T., et al., 1984, "A Method of Spatial Curve Tracing Control of an Articulated Robot," *Tr. SICE Jpn*, Vol. 20, No. 4, pp. 370–372 (in Japanese).

Reswick J. B., 1956, "Disturbance Response Feedback- A New Control Concept," *Trans. ASME* pp. 153–162.

Rosenbrock, H. H., 1974, *Computer-Aided Control System Design*, Academic Press, London, pp. 43–44.

Ryan, E. P., 1983, "A Variable Structure Approach to Feedback Regulation of Uncertain Dynamical System," *Int. J. Control*, Vol. 38, No. 6, pp. 1121–1134.

Seraji, H., 1987, "An Approach to Multivariable Control of Manipulators," *Trans. ASME*, Vol. 109, pp. 146–154.

Spong, M. W., 1986, "Robust Stabilization for a Class of Nonlinear Systems," *Theory and Applications of Nonlinear Control Systems*, Elsevier Science Publishers.

Downloaded From: http://dynamicsystems.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/journals/jdsmaa/26172/ on 06/14/2017 Terms of Use: http://www.asme.org/a